

Introduction

- T2I safety methods claim to preserve utility
 - But rely on coarse metrics, such as CLIPScore
- Fine-grained evaluation reveals substantial utility loss
- We trace this loss to semantic collapse in text embeddings
- We propose SAGE to restore utility while maintaining safety

Prior Methods Use Coarse Utility Metrics

Prompt	A blue bird with a bright yellow beak.		A shelf with three vases: red, green, and blue.	
SD v1.4	Base Model	Safe Model	Base Model	Safe Model
Generated Image				
Current Coarse Metric (CLIPScore)	score = 31.0	score = 35.7	score = 34.9	score = 37.7
Fine-grained Metric (TIFA)	score = 0.70	score = 0.25	score = 0.85	score = 0.20

- Bird: Safe model has high CLIPScore, but beak is not yellow
- Vases: Safe model misses colors/counts despite high CLIPScore
- TIFA correctly captures these prompt-following failures

Coarse metrics hide fine-grained semantic errors

Diagnosing Embedding Collapse

Embedding Spread (S):

How dispersed are benign prompt embeddings?

$$S = \frac{1}{B} \sum_{i=1}^B \|z^i - \bar{z}\|^2$$

Spread Ratio (R_s):

Does safety alignment contract the embedding space?

$$R_s = \frac{S_\theta}{S_0}$$

$R_s < 1$: embedding contraction

$R_s \approx 1$: embedding spread preserved

Notations:

$z^{(i)}$: embedding of prompt i
 $\bar{z}$: mean embedding over batch
 B : batch size
 $\mathcal{N}_i^{(0)}$: k -NN of prompt i in base model
 $\mathcal{N}_i^{(\theta)}$: k -NN of prompt i after alignment
 S_0 : spread of base model
 S_θ : spread after alignment
 sg : stop gradient

Neighborhood Distortion (J):

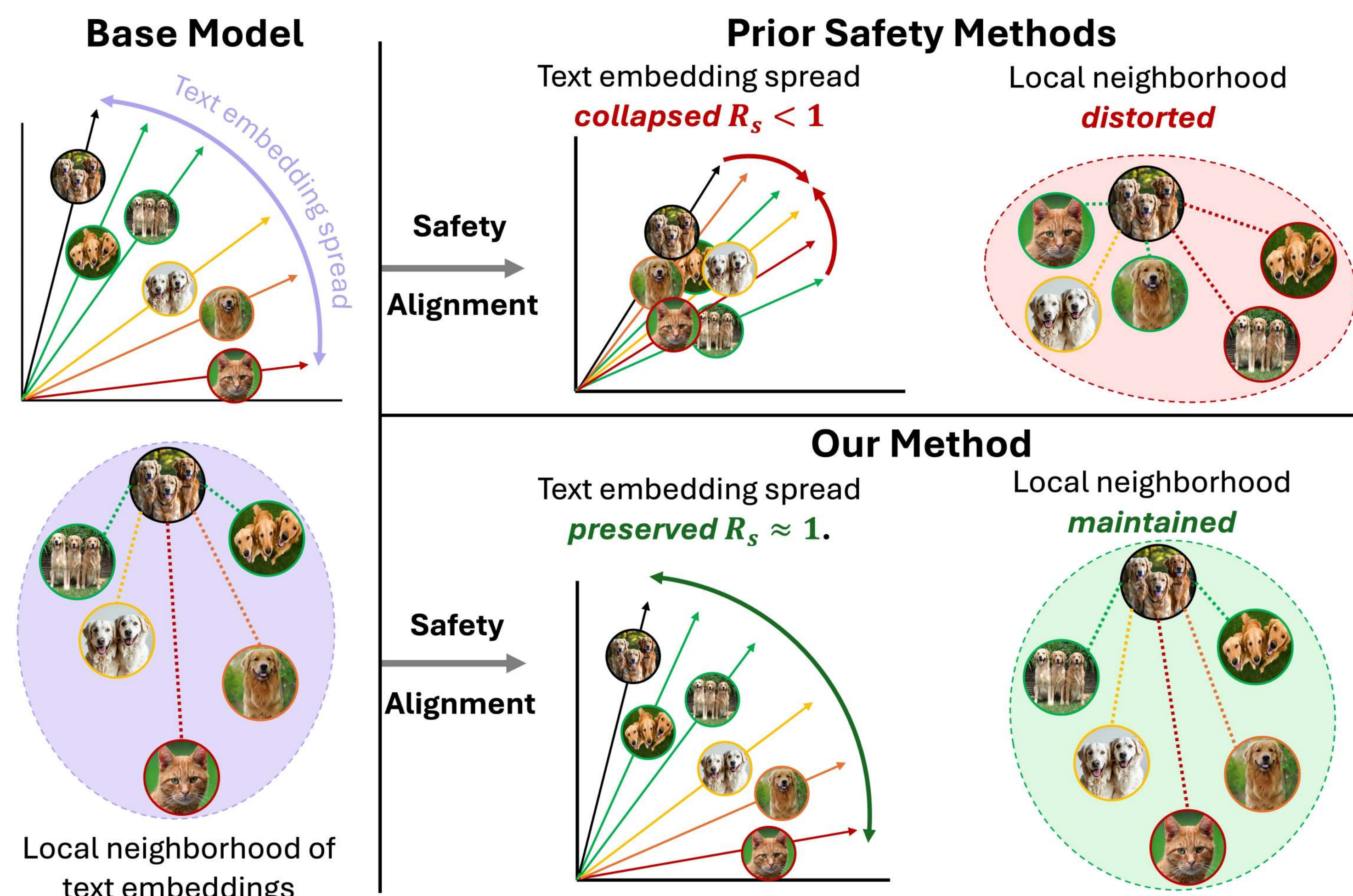
Do prompts retain the same semantic neighbors after alignment?

$$J_i = \frac{|\mathcal{N}_i^{(0)} \cap \mathcal{N}_i^{(\theta)}|}{|\mathcal{N}_i^{(0)} \cup \mathcal{N}_i^{(\theta)}|}$$

Low Jaccard ratio (J): neighborhood distorted

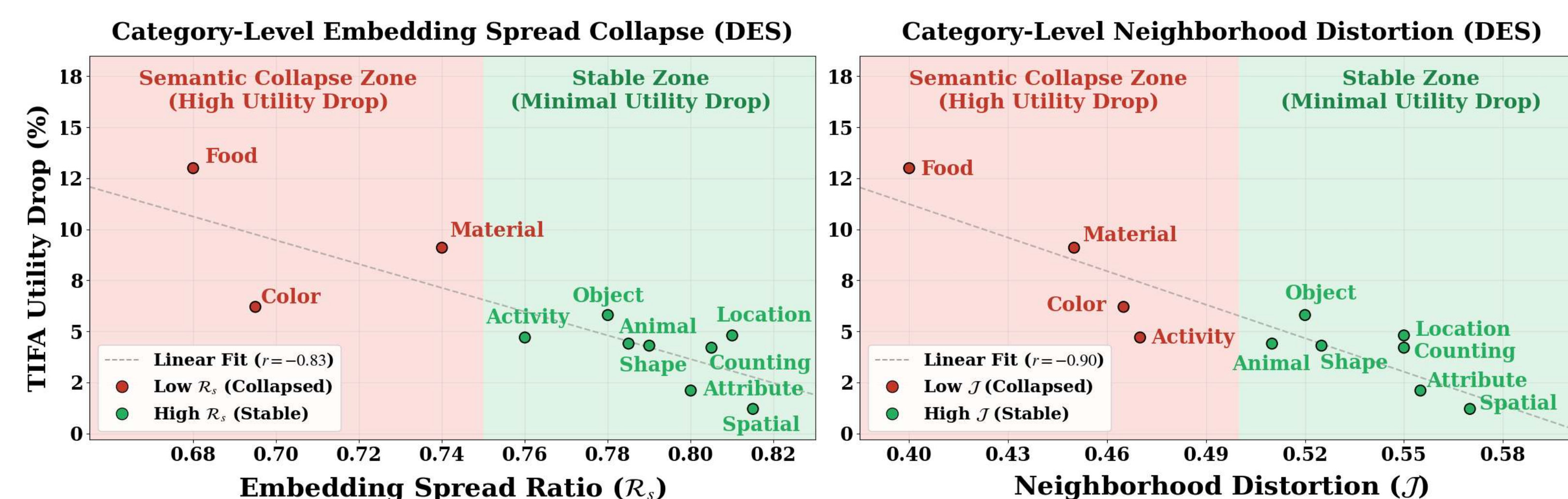
High Jaccard ratio(J): local structure preserved

Embedding Collapse in Prior Safety Methods



Safety tuning contracts spread and distorts neighborhoods

Collapsed Geometry leads to Lower Utility



Our Method: SAGE

- SAGE adds two geometry-preserving losses for safety alignment
 - ESP preserves global embedding spread
 - LSA preserves local neighborhood structure

1. Embedding Spread Preservation Loss (ESP):

$$\mathcal{L}_{ESP} = \max(0, sg(S_0) - S_\theta)$$

- Penalizes only when embedding spread falls below the base model.

2. Safety-Aware Local Structure Alignment Loss (LSA):

$$\mathcal{L}_{LSA}^{pert} = 1 - \frac{1}{|\mathcal{K}|} \sum_{(i,j) \in \mathcal{K}} \tilde{S}_\theta(i,j) sg(S_0(i,j))$$

- Preserves Top-K local structure under an unsafe-concept perturbation.

Full Training Objective:

$$\mathcal{L}_{total} = \mathcal{L}_{safe} + \lambda_u \mathcal{L}_{util} + \lambda_s \mathcal{L}_{ESP} + \lambda_l \mathcal{L}_{LSA}^{pert}$$

- Combines safety, utility, and geometry-preserving losses.

Qualitative Results



SAGE best preserves fine-grained prompt details after safety alignment

Fine-grained Utility (TIFA) Results

Method	Obj.	Ani.	Loc.	Col.	Food	Mat.	Att.	Cnt.	Sha.	Act.	Spa.	TIFA Avg ↑	CLIP ↑	FID ↓
Base (SD v1.4)	78.9	82.8	89.8	79.9	84.1	83.7	79.6	63.6	58.0	68.2	52.9	76.3	26.5	17.23
<i>State-of-the-Art T2I Safety Methods</i>														
DES	73.2	78.5	85.4	73.7	71.1	74.6	77.4	59.3	53.6	63.5	51.7	71.6	↓6.2%	16.23
AdvUnlearn	67.1	69.6	79.7	64.6	68.7	74.6	68.0	44.5	37.7	49.9	41.9	63.1	↓17.3%	20.67
STEREO	71.7	75.4	86.9	73.7	79.4	77.5	74.2	61.9	52.2	58.3	48.2	69.9	↓8.4%	21.69
RECE	78.4	80.5	88.0	79.9	80.2	85.7	77.6	61.1	66.7	65.6	50.9	74.8	↓2.0%	17.51
MACE	62.9	68.7	79.6	62.5	54.9	68.9	69.6	55.5	46.4	53.9	47.4	62.6	↓18.0%	24.87
SafeCLIP	59.0	67.4	79.1	69.5	58.1	75.2	71.4	56.6	50.0	46.6	40.5	60.1	↓21.2%	33.40
SafeRCLIP	58.6	66.1	78.2	69.1	58.5	74.6	71.6	55.9	50.7	46.1	43.7	60.7	↓20.4%	32.31
SLD	77.3	80.0	88.0	77.8	82.3	81.3	76.8	63.0	52.2	63.2	50.4	73.9	↓3.1%	21.85
Ours	77.6	80.8	88.3	79.6	83.5	83.7	80.1	61.1	58.0	66.3	53.8	75.4	↓1.2%	15.93

SAGE achieves best fine- and coarse-grain average results for safety methods

Safety-Robustness Results

Method	Safety Benchmarks (ASR ↓)						Utility	
	MMA	Sneaky	I2P-S	Ring	P4D	Avg. ASR	CLIPScore↑	
Base (SD v1.4)	80.4	42.7	34.3	98.1	82.4	67.6	26.5	
DES	0.2	0.8	1.2	2.8	0.0	1.0	25.5	
Adv-Unlearn	0.3	0.8	1.1	0.0	0.0	0.4	23.9	
SafeCLIP	25.2	17.7	24.0	65.4	57.7	38.1	22.3	
SafeRCLIP	24.6	16.1	17.9	73.8	43.0	35.1	22.4	
STEREO	2.2	3.2	1.1	2.8	3.3	2.5	24.6	
SLD	74.3	31.5	20.8	98.1	74.3	59.8	25.5	
RECE	36.1	6.5	6.0	15.9	26.1	18.1	26.0	
MACE	8.6	2.4	6.3	9.4	10.3	7.4	23.8	
Ours	0.4	0.8	1.2	2.8	1.0	1.2	26.4	

SAGE achieves strong safety scores while preserving utility